

Chapter 6: Non-Linear Spaces & Regularization

One concept, one exercise per slide — practice as you go

Concept: The Maximum-Margin Hyperplane

- An SVM finds the separating hyperplane with the WIDEST possible margin
- Primal objective: minimize $(1/2) ||w||^2$ subject to $y_i(w \cdot x_i + b) \geq 1$ for every point
- A wider margin means a more confident, more generalizable decision boundary



Exercise: Why does minimizing $||w||^2$ (rather than maximizing something directly) correspond to maximizing the margin?

Concept: Soft Margins and the Parameter C

- Real data is rarely perfectly separable — slack variables $\xi_i \geq 0$ allow margin violations
- Objective adds a penalty: $(1/2) ||w||^2 + C \cdot \sum \xi_i$
- Large C: few violations tolerated, narrower margin, more overfitting risk
- Small C: more violations tolerated, wider margin, more regularization



Exercise: You train an SVM and it overfits badly, achieving perfect training accuracy but poor test accuracy. Should you increase or decrease C?

Concept: The Dual Problem and the Kernel Trick

- The SVM dual depends on training data ONLY through pairwise inner products $x_i \cdot x_j$
- A kernel $K(x_i, x_j)$ can replace this inner product with one computed in a much higher-dimensional space
- The high-dimensional coordinates are NEVER actually computed — only the kernel values are
- This is how SVMs draw curved decision boundaries using only a linear-separator algorithm



Exercise: Explain why the kernel trick lets an SVM separate data that is not linearly separable in its original feature space, without ever explicitly transforming the data.

Concept: Mercer Kernels — Linear, Polynomial, RBF

- Linear kernel: $K(x,z) = x \cdot z$ — no transformation, a standard linear SVM
- Polynomial kernel: $K(x,z) = (x \cdot z + c)^d$ — curved, polynomial-degree boundaries
- RBF (Gaussian) kernel: $K(x,z) = \exp(-\gamma \|x - z\|^2)$ — infinitely flexible, local similarity



Exercise: A classic XOR-style dataset (two diagonal clusters of each class) cannot be separated by any straight line. Which kernel would you reach for first, and why?

Concept: Shannon Entropy and Information Gain

- Entropy $H(S) = -\sum p_k \log_2(p_k)$ measures how mixed a node's classes are
- $H=0$ means a node is perfectly pure (one class only); H is maximal when classes are balanced
- Information gain = parent entropy – weighted average of children's entropy
- A decision tree picks, at every node, the split that MAXIMIZES information gain



Exercise: A node has 8 examples: 4 of class A, 4 of class B. A candidate split produces two children, each perfectly pure (all A or all B). What is the information gain?

Concept: Why Decision Trees Overfit Without Limits

- An unconstrained tree keeps splitting until every leaf is perfectly pure
- This means it can memorize the training data completely, including its noise
- Regularize via `max_depth`, `min_samples_split`, `min_samples_leaf`, or pruning



Exercise: A decision tree trained with no depth limit achieves 100% training accuracy but only 65% test accuracy. What's the most direct first fix to try?

Concept: Random Forests — Averaging Away Variance

- Train many trees, each on a bootstrap-resampled subset of the data
- Each split considers only a random subset of features (typically $\sqrt{D}$)
- Averaging many decorrelated trees' predictions reduces variance without increasing bias much



Exercise: Why does training many trees on the SAME full dataset with the SAME features (no bootstrapping, no feature subsampling) fail to give the variance-reduction benefit of a real random forest?

Concept: L1 vs. L2 Regularization Geometry

- L1 (lasso) constraint region is a diamond with corners on the coordinate axes
- L2 (ridge) constraint region is a smooth sphere/circle
- The loss contours are more likely to touch a DIAMOND's corner — producing exact zeros
- A smooth sphere has no corners — coefficients shrink but rarely hit exactly zero



Exercise: Using the geometric picture (not the calculus), explain in one or two sentences why lasso can perform feature selection but ridge cannot.

Concept: SVM Margin Maximization Is Also Regularization

- SVM's $(1/2) ||w||^2$ term is structurally identical to L2 (ridge) regularization
- Margin maximization, decision-tree depth limits, and weight decay all share ONE principle:
- Penalize model complexity to prevent memorizing noise — the same idea, three different geometries



Exercise: Name the three chapter concepts (from this deck and Chapter 1) that are all instances of the exact same 'penalize complexity' principle, just applied in different model families.

Mastery Check: Are You Ready to Move On?

If you can answer every question below confidently, you have mastered Chapter 6.

1. Explain why minimizing $\|w\|^2$ is equivalent to maximizing the SVM's margin.
2. Explain the role of the parameter C and how it trades off margin width against violations.
3. Explain the kernel trick and why it avoids ever computing high-dimensional coordinates explicitly.
4. Compare linear, polynomial, and RBF kernels and when you'd choose each.
5. Compute Shannon entropy and information gain for a given node split by hand.
6. Explain why an unconstrained decision tree always reaches 100% training accuracy.
7. Explain why random forests need BOTH bootstrap sampling and feature subsampling to reduce variance.
8. Explain, geometrically, why lasso produces sparse solutions and ridge does not.
9. Explain why SVM margin maximization, ridge regularization, and tree depth limits are the same underlying idea.
10. Implement a simplified SMO solver or a recursive decision tree from scratch.